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Understanding Audiovisual Deepfake Detection: Techniques, Challenges, Human Factors, and Perceptual Insights

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Abstract—Deep learning has been successfully applied in various fields, and its impact on deepfake detection is no exception. Deepfakes are fake, yet realistic synthetic content

Digital Object Identifier 10.1109/MCI.2026.3656145
Date of current version: 7 April 2026

This article was recommended for publication by Associate Editor Alexander Wong. (Ammarah Hashmi and Sahibzada Adil Shahzad contributed equally to this work.) Corresponding author: Hsin-Min Wang (whm@iis.sinica.edu.tw).

that can be used deceitfully for political impersonation, phishing, slander, or the spread of misinformation. Despite extensive research on unimodal deepfake detection, the identification of complex deepfakes through joint analysis of audio and visual streams remains relatively unexplored. To fill this gap, this survey first provides an overview of audiovisual deepfake generation techniques, applications, and their consequences, and then provides a comprehensive review of state-of-the-art methods that combine audio and visual modalities to increase detection accuracy, summarizing and critically analyzing their strengths and limitations. Furthermore, we discuss existing open source datasets for a deeper understanding, which can contribute to the research community and provide necessary information for beginners who want to analyze deep learning-based audiovisual methods for video forensics. By bridging the gap between unimodal and multimodal approaches, this paper aims to improve the effectiveness of deepfake detection strategies and guide future research on cybersecurity and media integrity.

I. Introduction

The proliferation of smart digital devices such as mobile phones, laptops, tablets, and other digital gadgets, coupled with the accessibility of social media platforms, has promoted the exponential growth of multimedia content (images, videos, and audio) on the internet. This growth has been further fueled by technological advances [1], including various deep generative networks [2], [3], [4]. However, this accessibility increases the need for caution because it can lead to the prevalence of disinformation. Despite this issue, many people still adhere to the antiquated phrase “seeing is believing” and share multimedia content without considering its authenticity or verifying its digital integrity. Deepfake technology, which consists of sophisticated artificial intelligence (AI) models, enables deep learning (DL) tools to manipulate media (images, videos, and audio) to generate hyper-realistic fake content that deceives viewers. A deepfake is an AI-generated media that has been deceptively altered by superimposing a source face in a video onto a target face, manipulating the speech in an audio clip, or both. The vast amount of data available online in the form of images, videos, and audio to train such models makes detecting such forgeries increasingly challenging. The impact of deepfakes is critical because we still trust photographic and audio recording evidence, and the emergence of realistic and subtle production tools makes fake content incredibly believable and difficult to distinguish from genuine content [5]. The rapid spread of harmful and uncontrolled content from fake media has serious, imminent impacts and reduces trust in journalism and news providers [6], [7]. Deepfake media content can be exploited to fuel political or religious tensions between countries [8], spread misleading information or rumors between

political parties [6], [9], deceive the public [6], engage in revenge porn [9], defame celebrities [9], promote fraud and identity theft [10], and stir up political chaos or publicity in a campaign [11].

Generative adversarial networks (GANs) [2] and variational autoencoders (VAEs) [3] are sophisticated DL models for generating counterfeit content. In the GAN, the generator network and the discriminator network are the two main components, and these two networks oppose each other. The generator aims to generate plausible data, while the discriminator determines the real data from the fake data produced by the generator. Similarly, the VAE is an unsupervised learning method consisting of encoder and decoder architectures that is used to create high-quality, hyper-realistic fake content by merging and/or superimposing existing media (images or videos) onto source media for deception.

Currently, AI-synthesized videos can be divided into three main generation types [12], [13]. (1) Head puppetry/puppet mastery is a counterfeit video generation technique in which the target person is animated like a puppet. (2) Face swap aims to generate a video of the target person by swapping the target person’s face with that of the source person while retaining the same facial expression as the target person. (3) Lip-syncing is another deepfake video generation method where the main goal is to transform a person’s lip movements so that they are synchronized or consistent with the target audio. This technique tends to manipulate the lip region in such a way that the target appears to be saying things they never said in reality.

In the past few years, immense progress in the field of automatic video editing and great interest in facial manipulation techniques have been reported. Advances in manipulation tools and open-source codes allow even naive users to use deepfake technology like an expert in a few simple steps. This technological advancement has a wide range of positive applications in the fields of visual effects, photography, education, film, virtual reality, video games, cinema, and entertainment. However, it also poses significant challenges in terms of authenticity verification and prevention of malicious use. To overcome these challenges, researchers have made efforts in many studies and proposed DL-based unimodal forgery detection methods [14], [15], [16], [17], [18].

The detection of visual manipulation in videos has long been the focus of researchers, while the identification of audio forgeries has often been overlooked. However, recently, the trend of sound manipulation has grown rapidly along with visual alterations, leading to bimodal fabrication that enhances the authenticity of fake content and makes detection difficult [19], [20], [21], [22]. Figure 1 highlights the growing interest and concern of the research community with regard to audiovisual deepfakes. The number of publications on audiovisual deepfakes has increased significantly in recent years, demonstrating both beneficial progress and growing concerns.

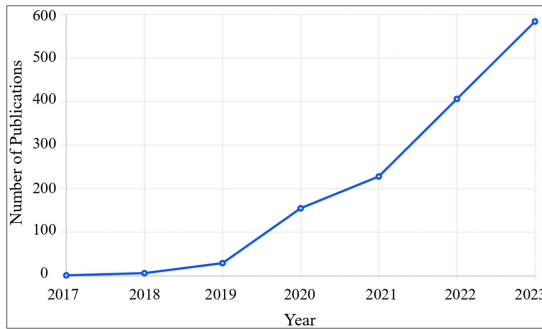


FIGURE 1 Volume of research into audiovisual deepfakes between 2017 and 2023.

The integration of multimodality has been proven to be beneficial in various research fields [23], [24], [25]. Consequently, researchers have used various DL techniques that exploit audio and visual features for video forgery detection. However, existing media forensics research lacks investigations that analyze methods for generating and detecting video deepfakes using audio and visual modalities. Table I lists an overview of the relevant studies. Our study is strongly motivated by the lack of attention paid to audiovisual deepfakes in surveys, which highlights the urgent need to focus research on audiovisual deepfakes, including their generation, how to mitigate their harmful effects, and a summary of existing audiovisual deepfake detection methods.

In general, as shown in Figure 2, there are four types of deepfakes, namely, text deepfakes, audio deepfakes, visual deepfakes, and audiovisual deepfakes. Audiovisual deepfakes

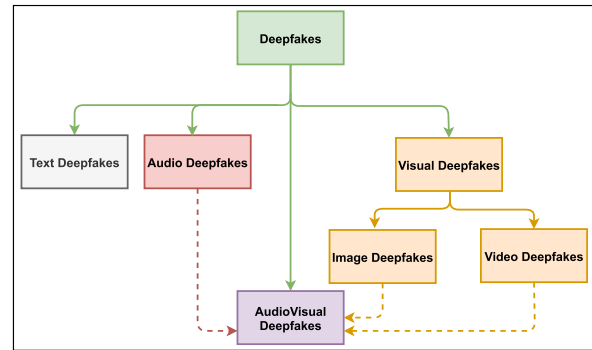


FIGURE 2 Taxonomy of deepfakes.

are a combination of acoustic and visual manipulation that can enhance manipulated videos to make them look more believable, and these approaches have received considerable attention in recent years. This study specifically provides an in-depth review of the latest audiovisual deep learning solutions to improve the detection of challenging video deepfakes. To the best of our knowledge, we are the first to perform a comprehensive analysis of existing DL-based methods that exploit audio and visual manipulations in videos for automatic deepfake detection. Our important contribution also includes a comprehensive discussion of publicly available datasets relevant to this task. The main contributions of our work are as follows:

- ❑ We provide an unprecedented survey that systematically analyzes key detection and generation methods for audiovisual deepfakes, with a special emphasis on automatic video deepfake detection methods.
- ❑ We highlight the challenges, limitations, and human perception in the field of audiovisual deepfake detection. We also outline research directions for future developments in this field.
- ❑ We summarize and present the publicly available datasets that can be used to train multimodal/audiovisual deepfake detectors.

The remainder of this paper is organized as follows: Section II introduces different types of deepfakes. Section III discusses deepfake video detection methods, and Section IV classifies detection methods that exploit visual and acoustic streams. In Section V, we review publicly available datasets for audiovisual deepfake detection. Section VI presents performance metrics and evaluation, and Section VII examines human perception of audiovisual deepfakes. Section VIII provides the limitations of the existing methods and datasets. Section IX further discusses several aspects of the deepfake challenge and potential research directions. Finally, Section X concludes this survey.

II. Deepfake Categories

Deepfake is a type of AI-generated fake hyper-realistic media content that involves the manipulation of acoustic and/or visual modalities, making it difficult to distinguish between

TABLE I Comparison of survey studies related to deepfake detection.

REFERENCE	YEAR	CONTRIBUTION
Verdoliva [26]	2020	A discussion of video deepfakes from a forensic perspective, with an emphasis on the limitations of current forensic detection methods.
Mirsky et al. [27]	2021	An in-depth analysis of field-specific generation techniques and a brief discussion of detection methods.
Yu et al. [28]	2021	A detailed analysis of fake video synthesis and detection techniques, with a focus on face manipulation.
Rana et al. [29]	2022	A comprehensive review of deepfake detection methods proposed during 2018–2020.
Nguyen et al. [30]	2022	A comprehensive overview of deepfake generation and detection techniques and a discussion of challenges and future research directions in the field.
Masood et al. [31]	2023	An analysis of the generation and detection of audio and visual deepfakes and a discussion of datasets.
Mubarak et al. [32]	2023	An analysis of audio, visual, and text-based deepfakes, with a focus on detection methods.

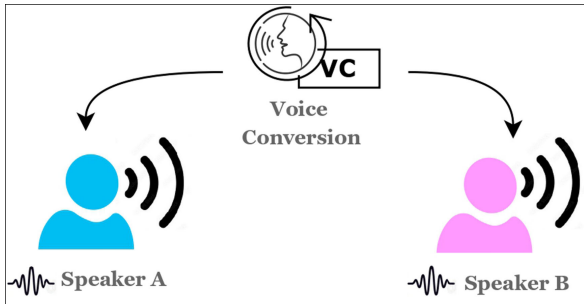


FIGURE 3 Diagram of voice conversion.

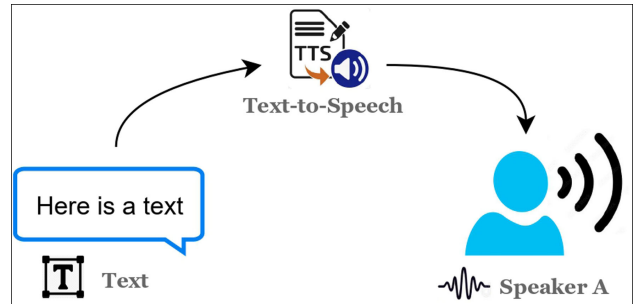


FIGURE 4 Diagram of text-to-speech.

real and fake media. This study focuses on audiovisual deepfakes, which combine audio and visual manipulations. In this section, we provide a brief overview of audiovisual deepfake generation methods and their components.

A. Audio Deepfakes

Audio deepfakes refer to voices that have been digitally altered but sound real. Convincing audio deepfakes are usually achieved through AI techniques or DL models, such as GANs or VAEs. MelGAN [33] and WaveGAN [34] are well-known examples. The former is a generative model for raw audio, and the latter is a generative adversarial network for conditional waveform synthesis. Contemporary breakthroughs in audio deepfakes have increased the threat to voice interfaces, contributing to criminal activity and cybercrime, thus raising concerns about their misuse. Audio deepfake methods fall into three categories, as described below.

1) Voice Conversion

Voice conversion (VC) involves converting the speech signal of a first speaker (the source) to match the voice of a second speaker (the target) while preserving the original linguistic content. Natural speech is input to the VC system, which aims to change the timbre and prosody of the source to that of the target. A schematic diagram of voice conversion is shown in Figure 3. Audio deepfakes can be used as independent clips for high impact, or they can be combined with visual deepfakes to produce audiovisual fabricated content. MelGAN-VC [35], an extension of MelGAN [33], is a voice conversion and audio style transfer method that can be applied to arbitrarily long samples. This method uses mel-spectrograms for voice conversion or style transfer through MelGAN. StarGAN [36] is a non-parallel many-to-many VC method based on star generative adversarial networks that can convert any source voice into the voice of any target speaker, and VoiceLoop [37] is a DL-based VC method that converts speech from the source to the target via a phonological loop. SINGAN [38] is a GAN-based singing voice conversion model that converts the singing voice of the source singer to the singing voice of the target singer.

2) Text-to-Speech (TTS)

Text-to-Speech (TTS), also known as speech synthesis (SS), involves the artificial generation of human speech with software or hardware systems. A schematic diagram of the TTS process is shown in Figure 4. TTS models typically follow two steps to convert written text into human-like speech: text processing and speech generation. This technology can be used to create fake audio messages or impersonate voices for threatening purposes, but it can also be useful for text-reading or personal assistants to provide accents and voices that are different from prerecorded human voices. Deep Voice [39] is a neural network-based synthesis model that generates text-to-speech in real time. Traditional TTS systems struggle to replicate natural flow and cannot mimic human speech. Recently, the quality of synthesized audio has been significantly improved due to the introduction of end-to-end models such as Variational Inference with Adversarial Learning for End-to-End Text-to-Speech (VITS) [40] and FastDiff-TTS [41]. Other DL-based methods include WaveNet [42], Deep Voice [43], Tacotron [44], and NaturalSpeech [45].

3) Partial Audio Deepfakes

In contrast to full audio deepfakes, partial audio deepfakes [46] involve the use of AI techniques to alter specific parts of the original clip while maintaining the overall authenticity of the recording rather than synthetically generating the entire audio. This complex selective alteration of specific segments of an audio recording makes detecting tampering more challenging than fully synthetic audio deepfakes. Figure 5 shows the generation process of a partial audio deepfake. Partial deepfakes of audio have a range of positive applications in entertainment and media, education, and accessibility, such as correcting errors in recordings; however, the technology's negative applications cannot be ignored, such as misinformation and disinformation, fraud and impersonation, and manipulation and defamation.

B. Visual Deepfakes

Visual deepfakes refer to images or videos that have been digitally altered by DL technology but look realistic. This technology can alter facial expressions, gestures, lip movements, and

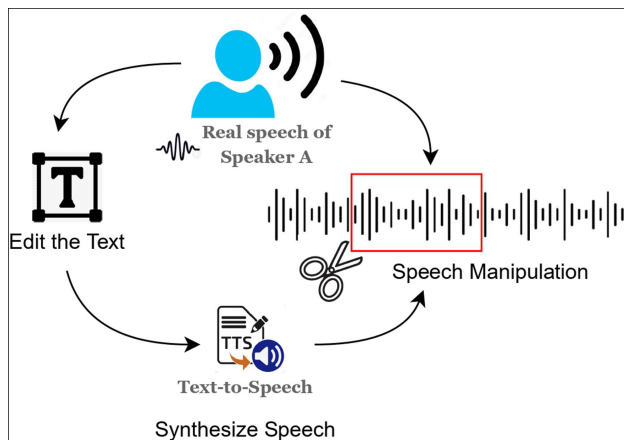


FIGURE 5 Schematic diagram of the generation of partial audio deepfakes.

body movements, or seamlessly superimpose the source person's face onto the target person's body and easily deceive the viewer. Although GAN-based methods dominate visual deepfake generation, recent advances in diffusion-based [47], [48] and autoregressive models [49], [50] have shown exceptional image synthesis capabilities. Diffusion models generate images through iterative denoising, offering improved sample diversity and stability compared to GANs. Autoregressive models generate images sequentially, capturing complex dependencies and providing an alternative to GANs. Visual deepfakes have widespread applications in fields such as education, gaming, and entertainment, but they pose considerable risks by causing misinformation, defamation, and privacy violations. Five types of visual manipulation fall into this category, each of which is discussed separately.

1) Face Swap

Face swap [51], a prominent visual manipulation technique, uses a generative model to seamlessly replace the target face with the source's identity in images or videos. An example generated using the face swap technique is presented in Figure 6. In certain scenarios, traditional methods [52], [53], [54] may fail to capture the expression depicted in the original facial image or sometimes result in an unnatural appearance. However, recent advances in DL have led to automated methods [55], [56] capable of producing plausible multimedia content. This technology has been integrated

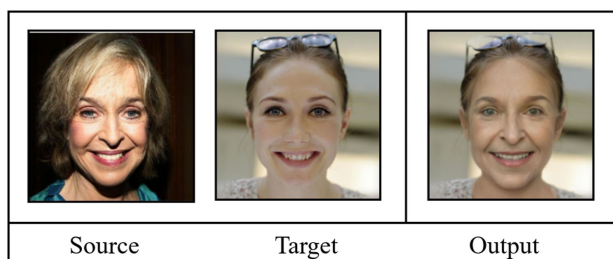


FIGURE 6 Face swap example [61].



FIGURE 7 Examples of facial generation.

into various applications, such as FaceSwap [56], DFaker [57], DeepFaceLab [58], DeepFake-tf [59], and FaceSwap-GAN [60], which use deep neural networks to achieve realistic face-swapping results.

2) Face Generation

Face generation, also known as face synthesis [62], creates realistic human faces that are completely fictional and do not correspond to any real identities using generative models such as GANs [2] and VAEs [3]. Some examples generated with this technique are shown in Figure 7. Although these images are entirely synthetic, their photorealistic quality and potential for misuse (e.g., in disinformation or fake profiles [63]) place them within the broader scope of deepfake technologies. This technology offers valuable applications, such as automatic character creation in the video game and 3D face modeling industries; however, it can also be used for malicious purposes, such as impersonating or spreading disinformation on social media [63]. Starting with the initial low-resolution imagery [2], the quality has improved dramatically over time. Advanced models such as StyleGAN [64], ProGAN [65], StyleGAN2 [66], TP-GAN [67], SAGAN [68], and BigGAN [69] can now produce high-resolution content.

3) Reenactment

Face reenactment, also known as puppet-master, is a technique for transferring facial expressions or body movements from a source video to a target video, as shown in Figure 8. Unlike face swap, this technique is designed to control and manipulate the target's expressions and movements to match those of the source, including gaze direction. Face reenactment has many applications in entertainment, virtual reality, and telepresence, but its potential for misuse may result in harm. Activists may create deceptive videos that manipulate a person's expressions and movements to falsely portray them as engaging in behavior or speech that never took place. Some publicly available tools that utilize tracking and reenactment techniques to demonstrate facial expression

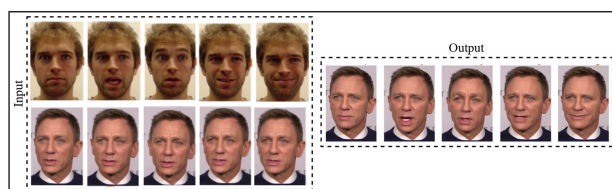


FIGURE 8 Reenactment example.

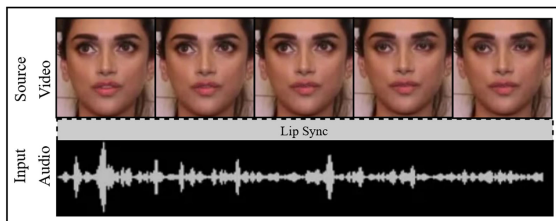


FIGURE 9 Lip sync example.

and motion transfer include Face2Face [70], FSGANv2 [71], MarioNETte [72], DeepFaceLab [58], pix2pixHD [73], and ReenactGAN [74].

4) Lip Sync

Lip sync involves changing a person's lip movements to match prerecorded audio clips, ensuring accurate and convincing speech alignment. An example of lip sync is shown in Figure 9. This process is complicated, as the appearance and movement of the lower face, lip region, and surrounding areas are crucial to achieve this goal, while precise lip movements and expressions are also important for effectively conveying a message. Lip sync has a variety of positive uses, including applications in forensic analysis, speech recognition research, and film production, where it ensures natural synchronization of the dialogue with the audio soundtrack. However, it can have negative consequences when it is used to generate deceptive content for malicious purposes, such as when it is used to spread false information, for defamation, or to manipulate public opinion. The authors in [75], [76], [77] proposed techniques for lip synchronization and other popular examples include Speech2Vid [78], Wav2Lip [79], LipGAN [80], and VDub [81].

5) Facial Manipulation

Facial manipulation aims to alter or synthesize facial features in images and videos to create realistic but counterfeit representations. This technology leverages advanced algorithms and DL techniques [14] to modify expressions, age, gender, and even the entire facial structure to create highly realistic or completely fictional portraits, as shown in Figure 10. It has positive applications in various fields, including entertainment, media, education, and training, but its negative potential raises significant ethical and security issues. FaceApp [82], SC-FEGAN [83], AttGAN [84], and OpenArt [85] are practical examples of manipulation tools that showcase advanced capabilities for altering and generating realistic facial images for entertainment purposes.

C. Text Deepfakes

Text deepfakes involve text that has been artificially generated or altered using advanced DL or AI techniques that produce highly realistic and contextually appropriate text. These sophisticated algorithms represent powerful and

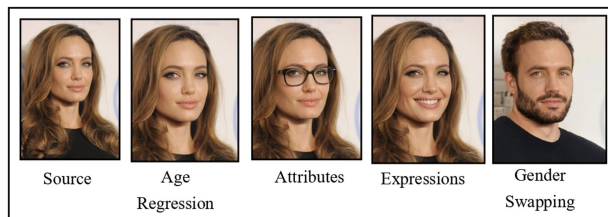


FIGURE 10 Examples of facial manipulation.

versatile applications of AI and are capable of generating articles, social media posts, conversations, and other forms of written communication that are often indistinguishable from human-written content. However, they can be misused; thus, a balanced approach that combines robust detection methods, ethical guidelines, and public awareness is needed. These deepfakes fall into two categories, each with unique characteristics. Both types are reviewed separately.

1) Synthetic Text

Synthetic text is generated by machines using advanced AI and DL techniques to imitate human writing with great accuracy. These types of text deepfakes are created using sophisticated natural language generation (NLG) models such as GPT-3, GPT-4, and their successors [86], [87], [88]. Large language models (LLMs) are trained on large amounts of human writing and generate coherent and contextual text based on a given prompt, making it difficult to discern between human and synthetic text [87], [88]. This technology has a variety of positive applications [86], including automated content generation for customer service, education, and creative industries. However, the potential for misuse, such as the spread of misinformation [89], fake news, and fraud, raises significant ethical and security concerns [90].

2) AI-Powered Bot-Generated Text

AI-powered bot-generated text refers to content created by automated systems or bots that use AI models to interact online. They excel at generating responses, posts, and other forms of communication that effectively mimic human behavior. AI empowers these bots to engage in social media discussions, comment on issues, and even generate news articles, often without being noticed by casual observers. The integration of AI into bots provides a variety of beneficial applications that increase efficiency and engagement on digital platforms. However, as technology evolves, the potential misuse of this deceptive approach poses significant challenges to the responsible use of AI in online interactions.

D. Audiovisual Deepfakes

Audiovisual deepfakes refer to convincing and compelling fabricated videos created by manipulating the acoustic and visual streams of a video, and bimodal manipulation produces sophisticated deepfake videos that seamlessly blend fabricated video

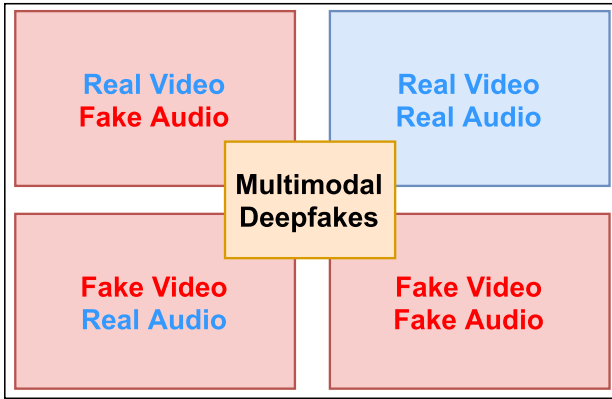


FIGURE 11 Four video categories based on acoustic and visual manipulations within them.

and audio content beyond single-modal manipulation. The misuse of audiovisual deepfakes poses significant risks, as they have the potential to spread misinformation and aid in defamation and other malicious activities. We discussed single-modal manipulation, namely, audio and video deepfakes, in the previous two sections. This section focuses on audiovisual deepfakes that combine the two, and categorizes the videos into four types based on the type of acoustic and visual manipulation, as shown in Figure 11.

1) Fake Video and Fake Audio (FVFA)

Unlike partial deepfakes, the Fake Video and Fake Audio (FVFA) category [91] involves the artificial generation of both audio and visual components. It is the most comprehensive and advanced audiovisual deepfake and is capable of fabricating complete acoustic and visual messages, often depicting fictional scenes that never happened. DL models, such as GANs, can generate high-quality synthetic visuals and produce realistic facial expressions, movements, and interactions that are coherent with the rest of the visual content. Moreover, synthetic audio that aligns with the video content is generated through TTS systems or voice cloning technology. These methods generate a highly persuasive audio track by analyzing the phonetic patterns of the target voice. An example of an FVFA is shown in Figure 12.

2) Real Video and Fake Audio (RVFA)

The Real Video and Fake Audio (RVFA) category [91] involves combining authentic video with synthetically generated audio recordings. An example of an RVFA is shown in

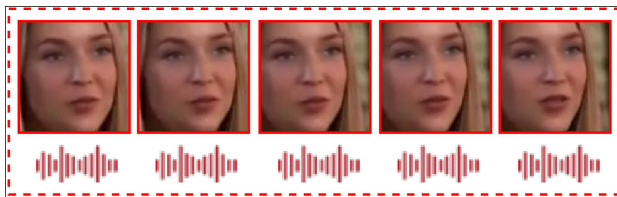


FIGURE 12 Example of the Fake Video and Fake Audio (FVFA) category.

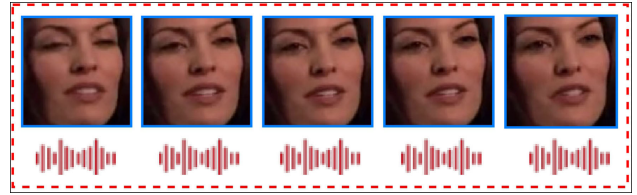


FIGURE 13 Example of the Real Video and Fake Audio (RVFA) category.

Figure 13. AI and DL techniques are used to alter auditory messages while preserving visual content, making people in videos appear to be saying things they never actually said. This approach is particularly effective in situations in which altering the audio can significantly change the perceived message or context of an event (such as political manipulation or the creation of fake news videos). The video is often based on real events or recordings and remains intact, while the synthesized audio is created using a voice cloning or TTS system to ensure alignment with the lip movements and expressions of the person in the video.

3) Fake Video and Real Audio (FVRA)

The Fake Video and Real Audio (FVRA) category involves [91] artificial manipulation of the visual track while the soundtrack remains real and intact, as shown in Figure 14. This category is particularly effective in situations in which altered visual scenes are combined with genuine audio to significantly change the perceived message or context of an event (such as in the creation of fake news videos, defamation, or impersonation). State-of-the-art visual methods are used to produce realistic visual effects that are consistent and synchronized with real audio to present a coherent and realistic audiovisual impression.

4) Real Video and Real Audio (RVRA)

The acoustic and visual tracks in the Real Video and Real Audio (RVRA) category [91] remain unaltered and are derived from authentic video and audio recordings, preserving the originality and integrity of both modalities. An example of an RVFA is shown in Figure 15. Technically, this category does not fall within the definition of deepfakes, but when used deceptively, it can be just as misleading as synthetic content.

III. Video Deepfake Detection Methods

The field of video deepfakes is thriving due to cutting-edge generative technology. At the same time, detection methods

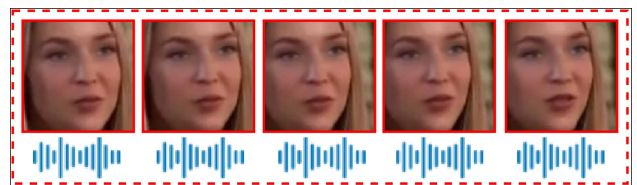


FIGURE 14 Example of the Fake Video and Real Audio (FVRA) category.

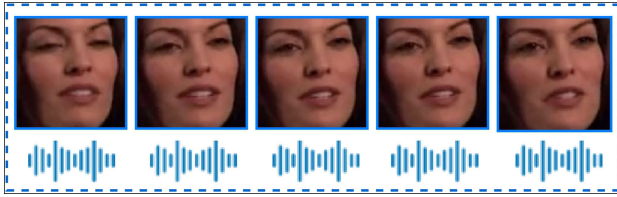


FIGURE 15 Example of the Real Video and Real Audio (RVRA) category.

have emerged as crucial tools for combating these deepfakes. This section explores various techniques for detecting video deepfakes, which can be divided into traditional methods based on visual cues and audio-based methods.

A. Traditional Methods Based on Visual Cues

Traditional methods for video deepfake detection rely heavily on visual irregularities or cues that arise while altering visual content [92], such as inconsistencies, anomalies, and artifacts within video frames [93]. These methods can identify subtle anomalies and are broadly divided into two subcategories, both of which are discussed separately below.

1) Frame Analysis

Frame analysis focuses on examining individual video frames to identify visual artifacts or inconsistencies such as unnatural lighting [94], irregularities in shadows [95], [96], and distortions around the edges of manipulated regions that indicate tampering [97]. Many techniques fall into this category, each focusing on different aspects of visual content. Pixel-level analysis [92], error level analysis (ELA) [98], compression artifacts [99], edge detection [100], and frequency domain analysis [100] are some examples of frame analysis techniques.

2) Inconsistencies in Facial Movements

Inconsistencies in facial movements are a major indicator of deepfake manipulation, which, along with expressions, are difficult to perfectly replicate. Conventional methods exploit these irregularities, such as unnatural or inconsistent facial movements [101], blinking patterns [102], eye movements [103], expression transitions [104], or synchronization issues, to spot deepfakes. Temporal consistency checks [105], landmark-based methods [106], blink rate analysis [107], and frame-by-frame analysis [108] are some examples of such techniques.

B. Audio-Based Methods

Audio-based deepfake detection methods emphasize analyzing audio aspects to reveal signs of tampering in videos. These methods are important traditional techniques for finding inconsistencies between audio and visual streams or detecting anomalies in the audio itself. Voice analysis and speech-content discrepancies are two main categories of audio-based methods to ensure the authenticity of audiovisual content.

1) Voice Analysis

Voice analysis techniques detect anomalies by analyzing acoustic properties such as speech patterns, voice characteristics, and linguistic nuances to identify deviations from natural audio characteristics [109], [110]. Analyzing audio frequency, pitch, and temporal characteristics provides valuable information for identifying fake multimedia content. Spectrogram analysis [111] and voice biometrics [112] are examples of voice analysis techniques.

2) Speech-Content Discrepancies

The alignment between the audio and visual components in deceptive content is often imperfect. When the movement of the lips does not align precisely with the words spoken, there can be a noticeable mismatch, indicating tampering. This technique uncovers inconsistencies that reveal manipulation by analyzing the synchronization between audio and visual components [18], [22]. Lip-sync detection [19] and Phoneme-Viseme Analysis [113] are well-known examples.

IV. Audiovisual Deepfake Detection Methods

Deepfake videos threaten personal privacy and social security. Various strategies have been proposed to overcome this risk, with early attempts focusing mainly on a single modality, whereas current detection methods focus more on multimodal features to identify manipulation in videos. Multimodal approaches combine the strengths of multiple modalities, such as audio and visual data, to increase the accuracy and reliability of deepfake detection systems. These methods can effectively discover more subtle manipulations by integrating information from both modalities that single-modal methods may ignore. This systematic literature review examines existing video deepfake detection methods that use both audio and visual streams to detect forgeries in videos. We categorize these methods, as summarized in Table II, and provide detailed information and discussion of the methods in each category later in this section.

A. Synchronization-Based Methods

Synchronization between audio and visual streams is a key aspect of multimodal deepfake detection and a prerequisite for creating a seamless viewing experience [114], [115]. These synchronization methods are designed to synchronize audio and visual streams in time, specifically lip movements, expressions, and overall timing. Since speech, lip movements, and facial expressions are naturally aligned in authentic videos, any discrepancy between audio and video streams indicates that the content has been tampered with or degraded, making synchronization techniques a critical tool in the fight against audiovisual fraud.

In [113], phoneme-viseme mismatches were exploited. The authors claim that producing sounds such as “M”, “B”, and “P” requires complete lip/mouth closure, while deepfake techniques often fail to imitate the dynamics of lip sequences,

TABLE II Comparison of audiovisual deepfake detection methods.

METHOD CATEGORY	KEY IDEA	STRENGTHS	LIMITATIONS
Synchronization-based	Exploits misalignment between audio and lip motion	Effective for detecting poorly synchronized fakes	Less effective when synchronization is realistic
Feature Fusion	Combines audio and visual features for classification	Captures multimodal correlations	Requires effective fusion strategies
Ensemble Methods	Uses multiple classifiers or modalities in combination	Improves robustness and generalization	Can be computationally expensive
Temporal Methods	Models temporal dependencies across frames/sequences	Good for detecting temporal inconsistencies	Sensitive to frame rate and sequence length
Other Methods	Includes anomaly detection, transformer-based models, etc.	Novel approaches; potential for high accuracy	May lack generalizability; often dataset-dependent

thus leaving clues that can be employed in forgery detection. In addition to modality-specific embeddings, [22] also integrated facial and speech emotion cues using state-of-the-art emotion recognition models to enhance the effectiveness of audiovisual forgery detection.

The authors of [21] exploited the inherent synchronization between video and audio streams to effectively identify inconsistencies and ultimately help spot manipulation in videos to improve deepfake detection. The proposed approach is a two-plus-one joint detection framework that jointly models audio and video, with one stream processing video frames and the other processing audio waveforms. The final deepfake prediction is produced by combining the outputs of these two streams, exploiting the inherent synchronization between them to detect misalignments between the two modes to identify forgeries in videos. Experiments demonstrate the high generalization ability of the proposed approach and show that the proposed joint audiovisual approach outperforms unimodal methods.

The authors of [116] exploited audiovisual dissonance, known as the Modality Dissonance Score (MDS), to expose video forgery. They reported that altering either modality can lead to noticeable dissonance, such as asynchronous lip movements or unnatural facial movements, which serve as indicators of altered material. Furthermore, their method provides a deeper understanding of altered video portions by locating segments in the video that exhibit signs of manipulation.

In [117], the limitations of previous studies that focused primarily on facial feature analysis and ignored the audio component and the broader context of audiovisual synchronization were addressed. The authors combined a phoneme-based audiovisual matching strategy and proposed the audiovisual coupling model (AVCM), which aims to capture the complex relationship between mouth movements (visual) and corresponding speech segments (audio) by measuring the similarity between them. Experiments show that their method performs well compared to cutting-edge techniques, highlighting its robustness in detecting deepfakes by leveraging the intrinsic synchronization of audio and video streams.

A unified audiovisual learning framework called AVoid-DF was proposed in [118], which effectively captures cross-

modal and intramodal inconsistencies or discrepancies between audio and visual content by jointly utilizing audio and visual features. This joint learning approach highlights the importance of using audio and visual cues simultaneously to substantially reduce the misclassification rate, especially in cases where one modality is tampered with.

The authors of [19] presented an interesting work that focused on the discrepancies between audio and visual components, specifically lip sync mismatch. Specifically, their proposed method detects inconsistencies between the lip sequence extracted from the video and the synthetic lip sequence generated from the audio using the Wav2Lip model. Furthermore, a pretrained convolutional neural network (CNN)-based lip-reading model is used to compare the extracted lip sequence with the synthetic lip sequence to distinguish synchronous and asynchronous audiovisual pairs, thereby helping to differentiate between real and fake videos. This work was further extended in [119], which utilizes audiovisual features extracted directly from the transformer-based multimodal model AV-HuBERT to identify inconsistencies between audio and visual components, thus eliminating the need for the Wav2Lip generation model.

B. Feature Fusion Methods

Feature fusion techniques leverage the complementary strengths of various features to enhance the accuracy of prediction models by combining features extracted from multiple modalities to create comprehensive and robust media representations. This holistic approach can significantly improve the accuracy of deepfake detectors, allowing them to capture more complex and informative representations of the underlying patterns.

In [120], the joint learning representation of audio and visual modalities was used to identify forgeries in videos, highlighting the importance of feature fusion. The authors proposed a specialized approach called Multimodaltrace, which leverages the rich interdependencies between audio and visual signals in video content. This framework efficiently evaluates the audio and visual modalities using a combination of channel extractors and mixers.

Another recent study in [121] proposed a dual transformer model called AVT2-DWF, which employs a dynamically weighted fusion strategy that calibrates its focus on each modality based on the relative strength and consistency of cues, thereby improving its ability to detect subtle or complex multimodal forgeries.

The authors of [122] proposed an innovative two-stage cross-modal learning method called audiovisual feature fusion (AVFF) to distinguish authentic video content from fabricated video content by focusing on audiovisual coherence. Their method identifies audio and visual perturbations by using distant audio and visual features and examining unimodal and cross-modal embeddings.

In [123], the authors introduced a novel one-class learning method that combines acoustic and visual features. This multi-stream fusion approach uses various fusion strategies (early, intermediate, and late) to effectively integrate audio and visual signals and surpasses single-modality detection models, demonstrating robust performance across diverse scenarios.

In [124], audio and visual modalities were utilized by extracting features and passing them to a multimodal model that learns nuanced differences across both domains. This framework explored two fusion strategies, feature fusion and score fusion, and demonstrated that simultaneously exploiting acoustic and visual features is effective in intradomain and cross-domain testing environments.

To address the limitations of heterogeneous feature fusion, a multimodal attention framework based on recurrent neural networks (RNNs) was proposed in [125] to exploit contextual information for audiovisual deepfake detection. The proposed approach uses attention mechanisms on multimodal, multi-sequence representations, identifying the most relevant features across modalities to improve deepfake detection and localization.

C. Ensemble Methods

Ensemble approaches have proven to be highly beneficial for improving the accuracy of detectors in terms of identifying audiovisual forgeries in videos, as they are able to take advantage of multiple models. Compared to a single model, integrating different models can effectively account for variations in forgery technology.

In [126], two ensemble methods for audiovisual deepfake detection were presented. The first method is soft-voting, which averages the probabilities predicted by each model; the second method is hard-voting, which applies a majority vote to the outputs of each model.

The authors of [127] introduced AVFakeNet, an ensemble architecture that uses visual and audio features to distinguish authentic videos from fake videos. Their model leverages a Dense Swin Transformer to handle bimodal complexity, extract and fuse audio and visual features, and ensure that the cross-modal relationship between visual and audio signals is effectively captured.

Similarly, [20] introduced an integrated network consisting of three CNN-based networks. The authors utilized ensemble

learning to integrate multiple modality-centric models to detect audiovisual forgeries in videos. Dedicated audio-only, video-only, and audiovisual networks make separate predictions, and a voting mechanism then fuses these predictions to make the final prediction. In the extension work, called AVTENet [128], the CNN architectures in audio-only, video-only, and audiovisual networks were respectively replaced by dedicated pretrained transformer models, thereby achieving better performance.

D. Temporal Analysis-Based Methods

To overcome the limitations of simple feature fusion techniques, the method proposed in [129] uses dual networks to extract temporal features from both modalities. Specifically, audio and video modules are used to predict acoustic and visual temporal features, which are then matched with reference features to capture temporal inconsistencies in audio and video modalities. Furthermore, a contrastive objective function is used to maximize the difference between authentic and spoofed modalities, significantly improving the model's ability to distinguish between real and fake instances.

For temporal forgery detection (TFD), a multidimensional contrastive loss was introduced in [130] to help forensic models exploit temporal inconsistencies by constraining extracted embeddings. Experiments on the LAV-DF [131] dataset demonstrate the effectiveness of the proposed method.

In [132], the authors proposed a hybrid approach that uses pretrained models to extract spatial, spectral, and temporal features to distinguish real content from fake video content.

A self-supervised learning (SSL)-based method was employed to capture rich representations based on temporal synchronization between speech and facial movement across frames in [133]. The representation learned by the SSL network is then input into a temporal classifier network to judge the authenticity of the video content.

E. Other Methods

Boundary-Aware Temporal Forgery Detection (BA-TFD) [131] was introduced to address temporal forgery localization tasks. It is a 3D convolutional neural network (3DCNN)-based model proposed to localize forgery through three guiding objective functions: contrastive, boundary matching, and frame classification. Later, BA-TFD+ [134] improved BA-TFD by replacing its backbone with a Multiscale Vision Transformer coupled with a refined training process that employs an additional multimodal boundary matching loss function.

MIS-AVoiDD [135] jointly utilizes modality-invariant and modality-specific features by focusing on shared and unique features across the modalities. The method proposed in [136] introduces a multimodal contrastive learning (MCL) approach to detect forgeries by capturing both intra-modal and cross-modal forgery cues. It aligns representations from audio, frames, and video using a cross-modal contrastive strategy and distills frame knowledge to the video network to strengthen forgery detection without extra computational cost. Additionally, a noise-based

TABLE III Comparison of existing audiovisual deepfake datasets.

DATASET	YEAR	MANIPULATION	SUBJECTS	REAL	FAKE	TOTAL
DFDC [142]	2019	AV	960	23,654	104,500	128,154
FakeAVCeleb [91]	2021	AV	500	500	19,500	20,000
LAV-DF [131]	2022	AV	153	36,431	99,873	136,304
AV-Deepfake1M [143]	2023	AV	2,068	286,721	860,039	1,146,760
PolyGlottFake [144]	2023	AV	-	766	14,472	15,238

feature augmentation (NFA) module further enhances generalization by adaptively perturbing audiovisual features.

DF-TransFusion [137] employs cross-attention between lip movements and audio signals to detect lip-sync inconsistencies, while a self-attention mechanism targets facial features to reveal subtle visual manipulations. This integrated approach improves detection accuracy by identifying cross-modal discrepancies, enabling more effective detection of complex deepfake manipulations.

In terms of real-world applications, [138] proposed a modality-agnostic-based method to address the missing modalities issue in the multimodal deepfake detection task. This method facilitates robust detection performance even when one of the input modalities, i.e., audio or video, is missing.

Another work [139] utilized monomodal datasets (visual-only or audio-only deepfakes) for training rather than relying on multimodal deepfake data. This approach involves extracting audiovisual features over time and analyzing them using time-aware neural networks, capitalizing on the inconsistencies both within and across modalities to improve detection performance. The proposed method was evaluated on unseen multimodal deepfakes, allowing it to evaluate the robustness of the detector without the need for multimodal training data.

To address the generalization issue in deepfake detection, a person-of-interest (POI) forgery detector was proposed in [140] to learn each individual's most discriminative identity features through a contrastive learning framework. This approach enables robust forgery detection without reliance on the artifacts or manipulation traces generated by deepfake techniques. Finally, [141] proposed a multimodal network with a web interface for real-time video forgery detection. This system integrates multiple data modalities to increase detection accuracy and provides a user-friendly, accessible platform for real-time analysis.

V. Datasets for Audiovisual Deepfake Detection

To effectively detect audiovisual deepfakes, a dataset with multiple manipulation methods, real-life scenarios, and increasingly sophisticated deepfake generation techniques is needed to provide a valuable resource for training, validation, and testing detection systems. This section discusses the existing widely used datasets that have been developed to aid in research of audiovisual deepfake detection. The statistics for these datasets are shown in Table III.

A. DFDC (DeepFake Detection Challenge)

A preview dataset containing 5,214 videos was released in October 2019, followed by the full dataset in December 2019. The full dataset contains 119,154 videos, each 10 seconds long, with 486 unique subjects; additional information is available in Table III. The DeepFake Detection Challenge (DFDC) [142] dataset is a large-scale publicly accessible dataset released by Facebook AI to promote research and development of deepfake detection methods.

This large-scale dataset was recorded in natural settings using high-resolution cameras without professional lighting or makeup and was collected from 3,426 paid subjects. It includes 1,000 deepfake videos generated using various techniques, including GAN-based and non-learning methods, for each genuine video. The dataset also includes various types of deepfakes, such as faceswap, face reenactments, and full-body deepfakes, and manipulations of audio, visual, or both audio and visual streams. This dataset highlights the diversity and variability of deepfake generation methods, focusing on audiovisual components, high-quality videos, and extensive use in benchmarking.

B. FakeAVCeleb

The FakeAVCeleb dataset [91] is a comprehensive collection of audiovisual deepfake recordings of celebrities designed to develop, advance, and evaluate deepfake detection methods. Developed by researchers at Sungkyunkwan University, South Korea, its primary purpose is to assist the scientific community in benchmarking deepfake detection models, particularly those that examine both visual and auditory modalities.

The dataset consists of 500 original videos, each approximately 30 seconds long, featuring a variety of celebrities from the sports, politics, music, and film industries, including Barack Obama, Donald Trump, and Kim Kardashian. Based on 500 original videos, 19,500 manipulated samples were generated using various manipulation techniques (such as FaceSwap [145] and Wav2Lip [79]) and real-time voice cloning (RTVC) (such as SV2TTS [146]). The dataset consists of 500 subjects; the videos for 470 subjects are used for training, and the remaining 70 are used for testing. The FakeAVCeleb dataset is known for its multimodal manipulation, which makes it more challenging and relevant than synthetic datasets that contain only unimodal manipulation. This challenging dataset can significantly contribute to academic research and practical applications in the fields of cybersecurity and media forensics and can be used to

benchmark deepfake detection, multimodal forgery detection, and human perception research.

C. LAV-DF (Localized Audio Visual DeepFake)

The Localized Audio Visual DeepFake (LAV-DF) dataset [131] is a multimodal dataset released by a group of researchers in 2022 to advance research on deepfake detection, with a particular focus on locating alterations in audio and visual data. It contains 136,304 videos, including 36,431 real videos of 153 unique subjects and the corresponding 99,873 fake videos. The LAV-DF features comprehensive audio and visual deepfakes with targeted content-driven manipulations (utilizing content-driven reenactment and text-to-speech methods) designed to more accurately replicate authentic speech and expressions. When evaluated on this dataset, the deepfake detection model needs not only to identify deepfake content, but also to accurately locate the fake areas.

D. AV-Deepfake1M

Released in 2023, the large-scale AV-Deepfake1M dataset [143] was carefully designed to facilitate the development of robust deepfake detection and localization techniques. The dataset contains more than 1 million videos featuring 2000+ subjects, encompassing audio, visual, and audiovisual manipulations. It provides a challenging benchmark for evaluating the efficacy of current detection models and encourages innovation to effectively mitigate the threat of deepfakes to media authenticity.

E. PolyGlottFake

PolyGlottFake [144] is a novel, comprehensive, multilingual, and multimodal dataset released in 2023, developed for advanced deepfake detection. PolyGlottFake spans seven languages, including English, French, Spanish, Russian, Chinese, Arabic, and Japanese, and incorporates a diverse set of manipulation techniques, including advanced text-to-speech, voice cloning, and lip-sync technologies. The dataset contains over 15,000 videos, encompassing both audio and visual manipulations, making it a valuable asset for advancing holistic deepfake detection research.

VI. Performance Metrics and Evaluation

Evaluating the performance of deepfake detection systems relies heavily on robust evaluation metrics that provide measurable insights into the model's accuracy, robustness, and generalizability, further allowing comparisons of various detection methods and progress observations. This section presents common metrics for evaluating audiovisual deepfake detection systems.

A. Accuracy-Based Metrics

1) True Positive Rate (TPR), False Positive Rate (FPR), True Negative Rate (TNR), and False Negative Rate (FNR)

Such metrics reflect a model's ability to correctly classify authentic content versus manipulated content. The true positive rate

(TPR), also known as the recall or sensitivity, reflects how effectively a model detects deepfakes and is defined as follows:

$$TPR = \frac{TP}{TP + FN}, \quad (1)$$

where TP and FN denote true positives (fake instances correctly detected as fake) and false negatives (fake instances incorrectly detected as real), respectively.

2) Precision, Recall, and F1-Score

Precision measures how accurately a model identifies fake instances. It is the ratio of TP to the sum of TP and FP :

$$\text{Precision} = \frac{TP}{TP + FP}. \quad (2)$$

Recall (also known as the sensitivity or TPR) evaluates the capability of a model to effectively identify all fake instances and is defined as follows:

$$\text{Recall} = \frac{TP}{TP + FN}. \quad (3)$$

Finally, the F1 score is the harmonic mean of the precision and recall:

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}. \quad (4)$$

This metric is particularly valuable in cases in which the dataset is imbalanced (when one category is more prominent than the other).

3) Accuracy

Accuracy is a widely used evaluation metric for classification tasks that provides an analysis of the overall performance of a model. It is the ratio of correct predictions (TP and TN) to the total number of instances in the dataset and is defined as follows:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}. \quad (5)$$

B. ROC and AUC

1) Receiver Operating Characteristic (ROC) Curve

The receiver operating characteristic (ROC) curve visually demonstrates how the model balances the true positive rate and the false positive rate by displaying the TPR versus the FPR under different thresholds. The ROC curve in the upper left corner represents the model with the highest TPR and the lowest FPR.

2) Area Under the Curve (AUC)

The area under the curve (AUC) refers to the area under the ROC curve. An AUC closer to 1 represents better performance, while an AUC of 0.5 indicates a lack of discrimination and is similar to random guessing.

VII. Human Perception and Psychological Impact in Audiovisual Deepfakes

Human perception plays a crucial role in spotting and analyzing the authenticity of audiovisual deepfakes. The complexity of human sensory processing allows individuals to detect subtle discrepancies, especially in nuances that automated systems may miss, such as facial expressions, tone of voice, lip synchronization, and synchronized audio and visual aspects. Human subjective judgment and detection proficiency can enhance the performance of automated detection systems.

A. Human Perception Studies in Deepfake Detection

1) Human Detection Capabilities and Challenges

Despite advances in audiovisual deepfake generation [91], humans inherently possess visual and auditory cognitive or perceptual skills that help detect subtle inconsistencies. However, studies in [147], [148], [149], [150] show that, contrary to expectations, a significant number of people are unexpectedly vulnerable to high-quality deepfakes, especially when subtle audiovisual alterations coincide with realistic behavioral signals [151].

Studies on the human ability to detect deepfakes [151], [152], [153], [154], [155] vary considerably due to factors such as the quality of manipulation, type of manipulation, familiarity with media figures, digital literacy, and age group [156]. Humans often rely on their own instincts, nonverbal cues such as unnatural eye blinks, discrepancies in lip synchronization, or inconsistencies in audiovisual synchronization [157] that advanced detection models may not capture.

2) Factors Influencing Human Detection Performance

Existing knowledge, cognitive biases, experience and expertise, contextual understanding, and emotional intelligence are several important factors that affect human perception and judgment of the authenticity of audiovisual content. Understanding these biases, the impact of prior knowledge, and having professional experience and expertise are critical to making nuanced decisions and are essential to improving the design of detection tools and their validity and reliability [154]. The uncanny valley effect, where near-perfect replicas appear perturbing and raise the uncertainty leading to deepfake detection, is particularly relevant for identifying deepfakes that almost, but not entirely, mimic real human behavior. However, humans often miss manipulation signs due to limited attention or fatigue, which is increased by prolonged exposure to digital content [153], [158].

B. Comparing Human and Machine Detection

1) Comparison of the Capabilities of Human and AI Models

AI algorithms for audiovisual deepfake detection surpass human perception in terms of detecting audiovisual deepfakes [151], [158], [159]. Unlike humans, these detection algorithms can analyze large amounts of content quickly, precisely,

and consistently. Many AI models are highly scalable and adaptable and free of biases that humans may have. However, humans possess unique abilities in terms of contextual understanding and subjective interpretation [154]. Social experience enables people to interpret nuances and adapt to complex, unforeseen scenarios. This helps them intuitively detect subtle irregularities in expressions or unnatural tones that may be lost without explicit feature engineering. This comparative strength highlights the promising benefits of integrating human and AI capabilities in practical applications [160].

2) Challenges in Automating Human Intuition

Translating subjective human intuition into automated objective, measurable features remains inherently difficult. The human ability to recognize micro-expressions, contextual inconsistencies, or subtle unnatural behaviors in manipulated content does not always translate into measurable features in algorithms, as this intuition is abstract. Some methods have demonstrated the potential of physiological data to bridge this gap, such as monitoring pupil dilation through eye tracking or measuring reaction times to audiovisual cues. Embedding insights from human intuition into machine learning frameworks may improve the accuracy of detection models [160].

C. Quantitative Evaluation of Human vs. Machine Detection Performance

A recent study [151] quantitatively compared human and machine performance in audiovisual deepfake detection using benchmark datasets such as FakeAVCeleb [91] and DFDC [142]. The results show that humans achieve moderate accuracy (around 64–66%), whereas state-of-the-art audiovisual forensic models such as LipForensics [18], AV-Lip-Sync [19], AV-Lip-Sync+ [119], CNN-Ensemble [20], and AVTENet [128] consistently outperform human judgment, achieving accuracies between 75% and 97%. These findings highlight two key observations. First, humans often rely on intuitive cues but struggle with subtle lip-sync artifacts, occlusions, or background noise, which makes detection particularly difficult in more diverse and uncontrolled datasets like DFDC. Second, while AI-based approaches demonstrate superior performance by leveraging multimodal consistency and advanced feature representations, their advantage is partly attributable to being trained on the same datasets used for evaluation. A more equitable comparison would therefore require assessing the generalization capabilities of both humans and models on unseen or in-the-wild data. Taken together, this evidence highlights the performance gap between human perception and automated systems, and underscores the need to develop models that complement human intuition in real-world detection scenarios.

VIII. Limitations of Current Methods and Datasets

Despite significant progress in audiovisual deepfake detection, existing methods exhibit several limitations that hinder their

robustness and generalization. Synchronization-based methods [21], [22], [113], [116], which rely on audiovisual lip-sync consistency, perform effectively under controlled conditions, with close-up frontal face videos and clean, single-speaker audio. However, their performance deteriorates when manipulations are subtle, lips or mouth are occluded, the face is distant from the camera, lip region extraction is challenging, speech is rapid, or the audio contains environmental noise or has been re-recorded. Feature fusion approaches [20], [126], [127], [128], which integrate audio and visual embeddings, generally achieve higher accuracy. However, fusion strategies are often simplistic, typically employing late feature or score fusion, which can fail to capture complex cross-modal interactions. In addition, these approaches tend to be visually dominant, with audio features contributing minimally, limiting their ability to detect subtle manipulations in either modality. Ensemble methods [20], [128] improve robustness by combining multiple individual detectors, but they require training each model separately and often involve computationally intensive preprocessing for each component model. This not only increases overall resource requirements but also makes the ensemble susceptible to inheriting the limitations of its individual experts and limits its ability to handle missing modalities. Temporal analysis-based methods [130], [132], [133], which exploit frame-to-frame dependencies, can capture cross-frame or cross-utterance inconsistencies, but are vulnerable to frame skipping, compression artifacts, and background noise, and require long input sequences, increasing inference costs. Overall, current methods often lack cross-dataset generalization and robustness to real-world distortions, limiting their applicability in practical forensic scenarios.

Existing datasets in audiovisual deepfake detection continue to face significant challenges. DFDC [142], despite its large scale and visual diversity, contains a limited number of audio forgeries and underrepresents multilingual and cross-cultural content, resulting in poor cross-dataset generalization. FakeAVCeleb [91], while pioneering in multimodal deepfake research, primarily consists of frontal face videos and lacks diversity in terms of video content. Models trained on this dataset often struggle to generalize to videos captured in uncontrolled, real-world environments, such as those found on social media platforms. Although the dataset includes some demographic diversity in terms of race, it remains heavily skewed toward English-speaking celebrities, limiting both linguistic and broader cultural representation. LAV-DF [131] contributes a large number of manipulated videos, but its manipulations are predominantly lip-sync based, restricting coverage of other forgery types. AV-Deepfake1M [143] provides substantial scale and modality diversity, yet its large size and the considerable computational resources required for preprocessing and training restrict accessibility. PolyGlotFake [144] enhances multilingual representation, but its overall size and range of manipulations remain modest compared to DFDC and AV-Deepfake1M. Collectively, these limitations demonstrate that while existing datasets have been instrumental in advancing detection methods, they still fall short in capturing the full

diversity and realism of manipulations encountered in real-world scenarios, highlighting the need for broader, more diverse, and more accessible benchmark resources.

IX. Current Challenges and Future Directions

Generalization remains a significant challenge for audiovisual deepfake detection. Existing audiovisual models perform well on seen datasets and deepfake techniques; however, their performance degrades significantly on unseen datasets and new manipulation techniques. Additionally, videos shared on online social media platforms often come with complexities, including illumination variation, non-frontal faces, far frontal faces, occluded faces or lips, background clutter, multiple speakers, compression artifacts, multiple camera angles, and environmental noise. These factors pose challenges to multimodal forgery detection systems, which are typically trained on clean, preprocessed, and biased datasets.

To address these generalization issues, it is crucial to develop diverse datasets that contain samples generated by a variety of state-of-the-art deepfake techniques and are free from gender and racial bias. The approach to dataset creation and development of multimodal deepfake detection systems must be iterative, and as new deepfake generation techniques emerge, it will be important to include more generated video samples in existing datasets. This continuous updating requires regular model revisions to effectively combat novel video forgery techniques. Furthermore, innovative approaches such as novel multimodal feature fusion and unsupervised or self-supervised learning-based foundation models can be leveraged to learn robust audiovisual features and potentially improve performance across datasets and manipulations. Techniques such as domain adaptation and transfer learning can further improve the generalization ability of multimodal deepfake detectors.

The scalability of current audiovisual forgery detection systems is also challenging. Deploying these systems on social media platforms, which host millions of videos and online users, requires computational efficiency to ensure real-time performance. Audiovisual forgery detection models process two data streams simultaneously, and this bimodal approach, combined with preprocessing and high data dimensionality in the forward pass, requires substantial resources to effectively analyze visual and acoustic features. To facilitate authenticity checking of visual and acoustic streams, lightweight video and audio feature extractors and resource-efficient neural networks are needed. For real-time detection systems, techniques such as pruning, quantization, and knowledge distillation must be employed to develop resource-efficient multimodal forensic methods. By optimizing these processes, we can improve the scalability and effectiveness of audiovisual forensic systems on dynamic online platforms.

Since deepfake manipulation primarily targets facial and speech data, serious privacy and ethical issues arise when collecting personal data to train data-driven end-to-end models. Consent must be ensured, and ethical standards must be adhered to, while strong privacy protections must be

implemented. These steps are critical for preventing potential harm and effectively addressing the challenges of forgery detection. Furthermore, it is crucial to incorporate model transparency and specific preprocessing steps before analyzing facial, lip, and speech biometric features. This approach not only enhances privacy protection but also ensures that individual rights are respected throughout the detection process. By prioritizing these ethical considerations, we can work toward developing more responsible and effective deepfake detection systems that protect individual privacy.

X. Conclusion

The rise of audiovisual deepfakes has eroded people's trust in digital content, causing seeing to fall short of believing. The ubiquity of audio and visual manipulation tools, coupled with the rapid and effortless spread of fake content through social media platforms, greatly increases the risk of abuse. In line with this trend, multimodality has also revolutionized the field of deepfake detection, showcasing cutting-edge performance and unparalleled versatility in terms of identifying forged content across different modalities. This survey highlights recent trends in multimodality in forgery detection and provides an in-depth look at the complexities and advances in multimodal or audiovisual deepfake detection technologies. We comprehensively explore existing studies, which use multimodality, especially audio and visual streams via AI algorithms, to detect forgery in videos. We also provide information on large-scale and high-quality datasets suitable for multimodal deepfake detection, as these characteristics are key factors in the advancement of detection methods. This survey also sheds light on the unique perspective of integrating human perceptual cues into algorithmic models, enabling these models to improve detection efficacy and strengthen defenses against audiovisual forgeries. We also delve into open issues and future directions to provide a roadmap for further research. This survey is expected to inspire researchers to incorporate multimodality into deepfake detection, providing them with important insights into how multimodality can contribute to multimedia forensic tasks.

Acknowledgment

This work was supported by the National Council of Science and Technology, Taiwan, under Grant NSTC 111-2221-E-001-002, Grant NSTC 112-2221-E-001-009-MY3, Grant NSTC 113-2221-E004-001-MY3, Grant NSTC 113-2221-E-004-006-MY2, and Grant NSTC 114-2221-E-001-006-MY3.

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